sorry i meant the message about how and where to start with my thesis

Ah, got it — you meant the *earlier message* (the detailed guide on how and where to start your thesis) in **English**, not the Python explanation.
No problem — here's a clear English version of that message

How to Start Your Thesis Project

Project title: *Biobased or Biodegradable Alternatives for Food Packaging: A Curated Dataset on Material Properties and Food Preservation Impact*

---

1. Understand the Final Goal

Your **end goal** is to create a **FAIR dataset** (Findable, Accessible, Interoperable, and Reusable) containing scientific data on **biobased and biodegradable packaging materials** and their **food preservation performance**.

You'll also produce:

1. A **systematic literature review** (PRISMA-based).
2. A **curated dataset** (Excel/CSV + metadata in JSON).
3. A **short paper** (Data in Brief style).
4. *(Optional)* A **dashboard** to visualize material properties and trends.

So, your project combines **data collection**, **data organization**, and **scientific analysis**.

🗓 2. Project Structure (6-Month Overview)

Phase	Months	Goal	Output
1. Orientation & Planning	1	Understand context, design search strategy	Research question + plan
2. Systematic Literature Review	1–2	Identify and select papers	PRISMA flow diagram + reference list
3. Data Extraction & Harmonization	2–4	Collect and structure material data	Draft dataset
4. Comparative Analysis	4–5	Analyze materials vs. conventional plastics	Graphs + insights
5. Reporting & Publication	5–6	Finalize dataset, paper, and visuals	Dataset + article + dashboard

---

🗓 3. Your First Steps

Step 1: Define Your Research Question

Example:

> "What are the physical, mechanical, and barrier properties of biobased and biodegradable materials used in food packaging, and how do these influence food preservation compared to conventional polyolefin-based plastics?"

Your research question guides what data you collect and which studies you include.

Step 2: Design Your Literature Search Strategy

Use scientific databases such as **Scopus**, **Web of Science**, or **Google Scholar**.
Possible search string:

("biobased" OR "biodegradable" OR "compostable") AND "food packaging" AND ("barrier properties" OR "mechanical properties" OR "shelf life")

Then apply the **PRISMA method**:

1. Identify: gather all search results
2. Screen: remove duplicates
3. Eligibility: read abstracts and exclude irrelevant ones
4. Include: select the final papers for data extraction

Step 3: Create a Data Extraction Template

Use Excel or Google Sheets. Include key columns like:

ID	Material	Chemical Origin	Processing Method	O_2 Permeability	Water Vapor Permeability	Tensile Strength	Elongation	Compatible Food Type	Shelf-Life Effect	Biodegradability	Reference

You'll fill this table with standardized information from each paper.
Later, you can export it as a **CSV + metadata JSON** for FAIR publication.

Step 4: Apply FAIR and Open Data Principles

Use metadata standards like **Dublin Core** or **DataCite**, including:

- Title, author(s), creation date, version
- Data description and variable definitions
- License (e.g., CC-BY 4.0)
- Persistent identifier (DOI) if published

This ensures your dataset can be reused and cited properly.

Step 5: Analyze and Visualize

Once the dataset is ready, analyze:

- Which materials have the best **barrier** and **mechanical** properties
- How these properties correlate with **food shelf life**
- How biobased materials compare to **conventional plastics (PE, PP, PET)**

Use visual tools (Excel, Python, R, or Power BI) to create:

- Boxplots or scatterplots of properties
- Trend charts of permeability vs. strength
- Summary statistics

Step 6: Final Outputs

You'll end up with:

1. **FAIR dataset** (Excel/CSV + metadata JSON)
2. **Data in Brief article** describing scope, structure, and potential uses
3. **(Optional)** Interactive dashboard or visual analytics report
4. **Full thesis** including introduction, methodology, results, and discussion

---

Summary: Where to Start

1. **Formulate your research question**
2. **Draft your literature search plan** (search terms + inclusion criteria)
3. **Create your data extraction template**
4. Then begin collecting and standardizing your data

If you'd like, I can help you next with:

- A **weekly timeline** for the 6 months
- A ready-made **Excel data extraction template**, or
- A **step-by-step PRISMA workflow**

Which one would you like to focus on next?

do you think there is room for creativity? what materials do i include?

Excellent questions — and I'm really glad you're asking them now.

Even though your thesis sounds very structured (FAIR dataset, systematic review, etc.), there's *definitely* room for **creativity** — both in *how* you design your dataset and *what* you choose to include. Let's unpack both parts:

---

1. Where Creativity Fits in This Thesis

Even in a data-driven, research-heavy project, you can shape **how the data tells a story**. Some ways to be creative:

📄 a. Design of the dataset itself

- You decide which **variables** to include — maybe you add novel ones like *renewable content (%)*, *CO_2 footprint per kg*, or *processing temperature range*.
- You can choose to organize your data in a way that's useful for a specific audience (researchers, packaging designers, policymakers).
- You could group materials by *feedstock type*, *biodegradability class*, or *end-of-life pathway* (e.g., compostable, recyclable, marine-degradable).

b. How you visualize or analyze the data

- Create **comparative plots** (e.g., barrier performance vs. fossil-based plastics).
- Build **clusters** or **material families** based on similar properties.
- Use a **dashboard** or infographic style to communicate your insights visually.

📄 c. How you connect materials to food applications

Instead of just listing properties, you can link them to real packaging cases:

> "PLA has low oxygen barrier performance, making it less suitable for fatty foods, but effective for short-shelf-life products like salads."

This interpretation adds creativity through **context and insight**, not just numbers.

📄 2. What Materials to Include

You want to cover **the main classes of biobased and biodegradable materials** used in *food packaging*. Below is a structured overview — think of these as "families" to include in your dataset.

A. Biobased, Biodegradable Polymers

Derived from renewable resources and biodegradable.

Category	Examples	Notes
PLA (Polylactic Acid)	From corn starch or sugarcane	Widely studied; brittle, low oxygen barrier
PHA / PHB / PHBV	From bacterial fermentation	Fully biodegradable, good barrier, limited thermal stability
Starch-based blends	Thermoplastic starch (TPS), starch-PCL blends	Biodegradable but sensitive to moisture
Cellulose derivatives	Cellophane, cellulose acetate	Excellent oxygen barrier; poor moisture resistance
Protein-based films	Gelatin, soy protein, whey protein	High oxygen barrier; often hydrophilic

---

📄 B. Biobased, Non-Biodegradable Polymers

Made from renewable sources but *chemically identical* to fossil-based plastics.

Category	Examples	Notes
Bio-PE / Bio-PP / Bio-PET	Made from bioethanol or sugarcane	Same properties as conventional versions; recyclable but not biodegradable
PEF (Polyethylene Furanoate)	From plant sugars	Promising PET alternative; excellent barrier properties

---

♻️ C. Fossil-Based, Biodegradable Polymers

Made from petroleum but still biodegradable — useful as comparison cases.

Category	Examples	Notes
PBAT (Polybutylene Adipate Terephthalate)	Often used in blends	Flexible and compostable
PBS / PBSA	Similar to PET alternatives	Biodegradable and thermally stable

---

📄 D. Natural or Composite Materials

Bio-based materials reinforced or combined with natural fibers.

Category	Examples	Notes

Category	Examples	Notes
Chitosan-based films	Derived from shrimp shells	Antimicrobial, oxygen barrier
Lignin-based composites	From wood waste	Enhances mechanical strength
Nanocellulose coatings	Paper-based packaging coatings	Strong, transparent, renewable

---

E. Conventional Fossil-Based Plastics (for comparison)

Include at least a few references for benchmark comparison:

- **LDPE / HDPE / PP / PET / PS / PVC**

These act as your “control group” when evaluating performance gaps.

Suggested Dataset Scope

To keep the project manageable, aim to include:

- **~30–50 unique materials or formulations**
- **~100–200 data points total** (barrier + mechanical + shelf-life data)
- Focus on **packaging-relevant studies** (films, trays, coatings, multilayers)

---

3. Balancing Creativity and Rigor

Your creativity should make the dataset:

- **More insightful** (link materials to food application or sustainability metrics)
- **More reusable** (clear structure, good metadata)
- **More engaging** (nice visual summary or dashboard)

In other words:

> Be creative in *how you curate and present* the data — but rigorous in *how you collect and document* it.

Would you like me to help you design:

- a **material classification framework** (how to group materials in your dataset), or
- a **creative visualization concept** (for your dashboard or final figures)?

Both are great ways to express creativity in this kind of scientific project.

i have to do this project for 24 weeks. can you make a weekly planning?

Absolutely — let’s build you a **24-week (6-month) thesis plan** that fits your project’s structure and deliverables.

This plan balances **literature work**, **data collection**, **analysis**, and **writing**, with flexibility built in for feedback and revisions.

24-Week Thesis Plan

Project title: *Biobased or Biodegradable Alternatives for Food Packaging: A Curated Dataset on Material Properties and Food Preservation Impact*

Phase 1 – Orientation & Setup (Weeks 1–4)

Goal: Understand the topic, define the scope, and prepare your tools.

Week	Focus	Main tasks	Deliverables
1	Topic deep dive	Read review papers on sustainable packaging, biobased polymers, and PPWR regulation	Short background summary
2	Define scope & research question	Write research question, objectives, and key definitions	Draft research question + project outline
3	Search strategy design	Build database search strings (Scopus, Web of Science, Google Scholar) and inclusion/exclusion criteria	Draft PRISMA protocol

Week	Focus	Main tasks	Deliverables
4	Project organization	Set up folder structure, reference manager (Zotero/Mendeley), and data template (Excel or Python script)	Ready-to-use data-extraction sheet

---

Phase 2 – Systematic Literature Review (Weeks 5–8)

Goal: Identify, screen, and select relevant studies.

Week	Focus	Main tasks	Deliverables
5	Literature search	Run database searches; export results (CSV, RIS)	Combined results spreadsheet
6	Screening stage 1	Remove duplicates, screen titles/abstracts	Updated PRISMA record
7	Screening stage 2	Read full texts, apply inclusion criteria	Final list of included studies
8	Documentation	Draw PRISMA flow diagram and summarize selection process	Literature review protocol section

---

Phase 3 – Data Extraction & Harmonization (Weeks 9–14)

Goal: Extract quantitative data and standardize it for your dataset.

Week	Focus	Main tasks	Deliverables
9	Initial extraction	Extract data from first 10–15 papers into template	Pilot dataset
10–11	Full extraction	Continue data collection; note units, conditions, and methods	Expanded dataset
12	Harmonization	Standardize units, fill missing metadata, ensure consistency	Clean, harmonized dataset v1
13	FAIR preparation	Create metadata (JSON or YAML); assign variable names and units	Draft FAIR metadata file
14	Quality check	Validate dataset (check ranges, outliers, duplicates)	Verified dataset v2

Phase 4 – Data Analysis & Visualization (Weeks 15–18)

Goal: Compare materials, interpret results, and visualize trends.

Week	Focus	Main tasks	Deliverables
15	Descriptive stats	Compute averages, ranges, correlations	Summary tables
16	Comparative analysis	Compare biobased vs. conventional plastics (barrier, mechanical, shelf life)	Analytical figures
17	Visualization	Create graphs (boxplots, radar plots, scatter plots)	Figures for thesis/paper
18	Interpretation	Link properties to applications and sustainability context	Results & discussion notes

---

Phase 5 – Writing & Publication Prep (Weeks 19–22)

Goal: Write the thesis and the *Data in Brief* paper draft.

Week	Focus	Main tasks	Deliverables
19	Thesis introduction & methods	Write background, methodology, and PRISMA section	Draft of Chapters 1–2
20	Results & discussion	Write results and interpretation sections	Draft of Chapter 3
21	Data in Brief article	Write and format dataset paper (scope, structure, reuse value)	Draft manuscript
22	Feedback & revisions	Review with supervisor, refine figures, check FAIR compliance	Revised versions

---

Phase 6 – Finalization & Defense Preparation (Weeks 23–24)

Goal: Final submission and presentation.

Week	Focus	Main tasks	Deliverables
23	Final editing	Proofread, finalize thesis layout, complete references	Final thesis manuscript
24	Submission & presentation	Submit thesis, prepare presentation slides/poster	Submission + defense slides

---

📋 Tips for Staying on Track

- **Weekly check-ins** with your supervisor help keep progress visible.
- Use **a shared folder** for dataset updates and literature PDFs.
- Keep a **research log** (even a short one-line per day) — it'll make writing your methods section much easier.
- Start **writing early** — don't wait until all data are complete.

Would you like me to make this plan into a **visual Gantt chart** (timeline view) or a **calendar-style checklist** that you can actually use for weekly tracking (e.g., in Excel or Notion)?

i want to make a powerpoint for my supervisor for next week to update her on my progress and questions that i have. how shall i structure it?

Perfect timing — doing an early progress presentation is an excellent move. It shows initiative and helps your supervisor guide you *before* you go too deep in the wrong direction.

Here's a **recommended PowerPoint structure** (8–10 slides) that's concise, professional, and specifically tailored for your thesis topic:

PowerPoint Structure: "Thesis Progress & Next Steps"

Slide 1 — Title Slide

Purpose: Set the scene.
Content:

- Thesis title

  > *Biobased or Biodegradable Alternatives for Food Packaging: A Curated Dataset on Material Properties and Food Preservation Impact*

- Your name, degree program, supervisor's name
- Date of presentation

Slide 2 — Background & Motivation

Purpose: Remind your supervisor of the broader context.
Content:

- Short summary of the sustainability challenge (plastic waste, PPWR regulations)
- Why sustainable packaging materials matter
- Key gap: *lack of centralized, standardized data on biobased/biodegradable material properties*

Visual idea: a simple diagram showing "Conventional plastics → Sustainable alternatives → Data gap".

Slide 3 — Project Aim & Objectives

Purpose: Make your goals explicit.
Content:

- **Main aim:** Develop a FAIR dataset on biobased/biodegradable packaging materials and their food preservation performance.
- **Objectives:**
  1. Conduct a systematic literature review (PRISMA-based).
  2. Extract and harmonize data on physical, mechanical, and barrier properties.
  3. Compare materials to conventional plastics.
  4. Publish dataset + Data in Brief article (+ optional dashboard).

Visual idea: Use an icon timeline or four-block layout (one block per objective).

Slide 4 — Research Question

Purpose: Clarify your scientific direction.
Example content:

> "Which biobased and biodegradable materials show suitable mechanical and barrier properties for food packaging applications, and how do these properties impact food preservation compared to conventional polyolefins?"

Add your sub-questions if you've defined any (e.g., "Which material classes perform best under humid conditions?").

Slide 5 — Methodology Overview

Purpose: Show your workflow and reassure your supervisor you have a structured approach.
Content:

- PRISMA-based systematic review
- Data extraction & harmonization
- FAIR dataset preparation
- Comparative analysis

- Visualization & publication

Visual idea: a flow diagram with 5 boxes (input → process → output).

Slide 6 — Work Completed So Far

Purpose: Show concrete progress.
Content:

- Background reading (list a few key review papers you've read)
- Defined research question and scope
- Drafted PRISMA search strategy
- Designed preliminary data-extraction template
- Set up folder structure and reference manager

Visual idea: checklist or progress bar (e.g., "Phase 1 – 80% complete").

Slide 7 — Preliminary Insights or Challenges

Purpose: Start discussion and show you're thinking critically.
Examples:

- Some materials have inconsistent property units (need harmonization method).
- Many papers lack full food shelf-life data.
- Decision: whether to include *bio-based non-biodegradables* (Bio-PE, Bio-PET) as a separate category.

Visual idea: two columns → "Observations" vs. "Questions/Challenges".

Slide 8 — Next Steps (Next 4–6 Weeks)

Purpose: Show planning and momentum.
Content:

- Finalize literature search & screening (Weeks 5–8)
- Begin data extraction
- Start building FAIR metadata structure
- Plan short meeting for feedback before dataset analysis

Visual idea: simple Gantt-style mini-timeline.

Slide 9 — Key Questions for Discussion

Purpose: Guide your supervisor's feedback.
Example questions:

- Should the dataset include **bio-based non-biodegradable plastics** (e.g., Bio-PET)?
- How deep should I go into **food shelf-life parameters** (microbial data, oxidation)?
- What **metadata standard** would be most suitable for FAIR structure (Dublin Core vs. DataCite)?
- Is a **visual dashboard** expected or optional?

Slide 10 — Closing Slide

Purpose: End clearly and professionally.
Content:

- Thank you / open for feedback
- Your contact details (if relevant)
- Optional: inspirational quote or summary statement (e.g., "Turning data into sustainable design insights").

🗒 Extra Tips

✅ **Keep it visual** – Limit text, use diagrams for workflow and scope.
✅ **Keep it short** – Aim for **7–10 minutes** of talking + discussion time.
✅ **Show curiosity** – Supervisors love to see thoughtful questions, not just completed work.
✅ **Bring your data template** – You can add one example slide with your Excel layout or variables list (optional).

If you want, I can create a **PowerPoint outline (titles + bullet text)** you can copy directly into slides — or even a **slide-by-slide draft script** for what to say on each slide.

Would you like me to draft that next?

how does prisma search work? can you make a step guide?

Absolutely — this is a great question, because understanding how **PRISMA** works is essential to making your thesis *systematic* and *credible*.

Let's go step-by-step through what PRISMA actually means, how you apply it, and how you can document it for your project.

What PRISMA Is

PRISMA stands for **Preferred Reporting Items for Systematic Reviews and Meta-Analyses**.

It's a standardized method for conducting and *reporting* systematic literature reviews — ensuring that your process is **transparent, reproducible, and unbiased**.

In your case, you'll use it to **identify, screen, and select** studies that contain data on **biobased or biodegradable food packaging materials**.

🧾 PRISMA Workflow — Step-by-Step Guide

STEP 1: Define Your Research Question

Everything starts here.

Example for your thesis:

> "Which biobased or biodegradable materials are used for food packaging, and what are their mechanical, barrier, and preservation properties compared to conventional plastics?"

This defines **which studies** are relevant and which are not.

✅ *Tip:* Use the **PICO framework** (Population, Intervention, Comparison, Outcome):

- **P:** Packaging materials used for food contact
- **I:** Biobased or biodegradable materials
- **C:** Conventional plastics (PE, PP, PET)
- **O:** Material properties & food preservation performance

STEP 2: Identify Databases and Search Terms

Choose your sources (usually 2–3 scientific databases):

- **Scopus**
- **Web of Science**
- **Google Scholar** (for additional coverage)

Then build your **search string** using Boolean operators:

```
("biobased" OR "biodegradable" OR "compostable" OR "bio-based polymer")
AND "food packaging"
AND ("barrier properties" OR "oxygen permeability" OR "mechanical properties" OR "shelf life")
```

✅ *Tips:*

- Use filters: *journal articles*, *English*, *2010–2025*.
- Save/export your search results (title, authors, abstract, DOI).

STEP 3: Identification (Collecting All Possible Studies)

- Run your search in all databases.
- Export all results (usually as .CSV or .RIS files).

- Combine them in Excel or a reference manager (e.g., **Zotero**, **Mendeley**, or **Rayyan**).

You'll probably have **hundreds of hits** — many duplicates.

✅ *Deliverable:*
A big spreadsheet with all found studies and a column for "Database source".

STEP 4: Screening (Remove Duplicates + Title/Abstract Review)

Now you narrow down the list.

Phase 1 – Deduplication
Remove all duplicate papers (same title or DOI).

- Tools: EndNote, Mendeley, or Python/Excel filter.

Phase 2 – Title & Abstract Screening
Read titles and abstracts to check relevance.
Exclude:

- Non-food applications (e.g., bioplastics for agriculture).
- Papers without material property data.
- Non-English papers.

✅ *Deliverable:*
Spreadsheet with columns like "Include/Exclude" and "Reason for exclusion".

STEP 5: Eligibility (Full-Text Review)

Read the full text of the remaining papers.
Include only those that **report quantitative data** on:

- Mechanical or barrier properties, and/or
- Food preservation or shelf-life effects.

Document your decisions:

Paper ID	Include?	Reason
1	Yes	Contains O_2 and H_2O permeability data for PLA
2	No	Only reviews; no primary data

✅ *Deliverable:*
A "final included studies" list (probably 30–60 papers).

STEP 6: Inclusion (Final Dataset Source List)

You now have the **final selection** of studies that will feed into your dataset.

Extract data from these papers into your Excel sheet with fields like:

- Material type
- Processing method
- O_2, CO_2, and water vapor permeability
- Tensile strength, elongation
- Food shelf-life indicators
- Biodegradability
- Reference (DOI)

✅ *Deliverable:*
The *included studies table* (the foundation for your dataset).

STEP 7: Document with a PRISMA Flow Diagram

Finally, you summarize your process visually.

It usually looks like this:

Records identified through database searching (n = 500)

↓
Records after duplicates removed (n = 420)
↓
Records screened (titles/abstracts) (n = 420)
↓
Records excluded (n = 350)
↓
Full-text articles assessed for eligibility (n = 70)
↓
Full-text articles excluded (n = 25)
↓
Studies included in final dataset (n = 45)

✅ *Deliverable:*
A PRISMA flow diagram (use the official template: https://prisma-statement.org/)

You'll include this figure in your thesis and Data in Brief paper.

STEP 8: Transparency & Reproducibility

Keep a record of:

- The exact date and version of your searches
- Database names and filters used
- Your inclusion/exclusion criteria

This ensures your dataset is **FAIR and reproducible** — a key requirement for publication.

📋 Summary Checklist

Step	PRISMA Stage	Your Output
1	Define research question	Clear scope & keywords
2	Search databases	Search string + export results
3	Identification	Master list of all hits
4	Screening	Shortlisted papers (title/abstract)
5	Eligibility	Full-text selection
6	Inclusion	Final dataset source list
7	Documentation	PRISMA flow diagram
8	Transparency	Appendix with methods

Would you like me to make you a **PRISMA Excel template** (for tracking papers and reasons for inclusion/exclusion), or a **visual PRISMA flow diagram** example based on your project numbers?

so you have other Mnemonics for structuring questions?

Excellent follow-up — yes!

PRISMA often works best when your **research question** is crystal clear, and the **PICO** structure I mentioned earlier is just *one* of several mnemonics you can use to frame it systematically.

Depending on your research **type** (quantitative, qualitative, mixed, environmental, etc.), different mnemonics help define the right scope and variables.

Here's a breakdown of the most useful ones

📘 1. PICO – The Classic for Quantitative / Experimental Questions

Best for: Comparing interventions or materials (like your project).

Element	Meaning	Example (your thesis context)
P	Population / Problem	Food packaging materials
I	Intervention	Biobased or biodegradable materials
C	Comparison	Conventional plastics (PE, PP, PET)

Element	Meaning	Example (your thesis context)
O	Outcome	Mechanical, barrier, and food preservation properties

✅ **Example question:**

> "How do biobased and biodegradable packaging materials compare to conventional plastics in terms of barrier performance and food preservation?"

---

2. PEO – For Observational or Environmental Questions

Best for: Exploring associations, not testing interventions.

Element	Meaning	Example
P	Population / Problem	Food packaging systems
E	Exposure	Use of biobased or biodegradable materials
O	Outcome	Impact on food preservation and sustainability

✅ **Example question:**

> "What is the effect of using biodegradable packaging on the shelf life of perishable foods?"

---

3. SPIDER – For Qualitative or Mixed-Methods Research

Best for: When you're exploring perceptions, opinions, or experiences (e.g., industry or consumer views).

Element	Meaning	Example
S	Sample	Food packaging industry / research community
PI	Phenomenon of Interest	Adoption of biobased packaging materials
D	Design	Literature review / interviews / surveys
E	Evaluation	Perceived performance and sustainability
R	Research type	Qualitative / quantitative / mixed

✅ **Example question:**

> "How do packaging designers and researchers evaluate the performance and feasibility of biobased packaging materials?"

---

4. SPICE – For Policy, Contextual, or Applied Questions

Best for: Evaluating the *effect* of a change in a real-world context (fits sustainability studies well).

Element	Meaning	Example
S	Setting	Food packaging sector in the EU
P	Perspective	Regulators / packaging manufacturers
I	Intervention	Transition to biobased packaging
C	Comparison	Traditional fossil-based packaging
E	Evaluation	Compliance with EU PPWR & environmental performance

✅ **Example question:**

> "How does the adoption of biobased packaging materials in the EU food sector compare to traditional plastics in meeting PPWR sustainability targets?"

---

5. PICo – For Broad or Exploratory Reviews

Best for: Scoping reviews or non-experimental systematic reviews.

Element	Meaning	Example
P	Population	Food packaging materials
I	Interest	Biobased or biodegradable alternatives
Co	Context	Food preservation and circular economy goals

✅ **Example question:**

> “What biobased or biodegradable materials have been developed for food packaging, and how do their properties support circular economy goals?”

📋 Choosing the Right Mnemonic

Research Type	Recommended Mnemonic
Quantitative comparison (like yours)	**PICO**
Environmental or exposure study	**PEO**
Policy or applied sustainability	**SPICE**
Broad mapping or scoping review	**PICo**
Qualitative or perception-based	**SPIDER**

✨ For Your Thesis

Since your work is **data-driven, comparative, and systematic**, the best fit is:

> **PICO**, possibly extended with a **“T” for Timeframe** (→ **PICOT**) if you consider storage duration or shelf-life periods.

So, your refined version could be:

Element	Description
P	Food packaging materials
I	Biobased or biodegradable alternatives
C	Conventional fossil-based plastics
O	Physical, mechanical, and barrier properties; food preservation performance
T	Storage duration / shelf life of packaged food

✅ **Example question:**

> “Over typical storage durations, how do biobased and biodegradable food packaging materials perform compared to conventional plastics in terms of mechanical and barrier properties and their impact on food shelf life?”

Would you like me to help you **turn your question into a PRISMA search strategy** (keywords + Boolean operators + inclusion criteria) using one of these frameworks (like PICO or PICo)?

what type of research question will i have? : descriptive, comparative, evaluative, defining, explaining or designing

Excellent question — and this is exactly the kind of conceptual clarity your supervisor will love seeing in your presentation.

Let’s unpack it carefully and match your **project description** to the **type of research question** you’re actually addressing.

🧭 Step 1: Revisit Your Aim

Your thesis aims to:

> “Create a FAIR dataset on biobased and biodegradable food packaging materials by systematically collecting and harmonizing data on their physical, mechanical, and barrier properties, and their impact on food preservation — and compare them to conventional plastics.”

So you are not inventing a new material (→ not *designing*), and you’re not testing hypotheses in a lab (→ not *explaining* in a causal sense).

Instead, you are **mapping, comparing, and evaluating** existing knowledge to make it accessible and reusable.

🧭 Step 2: Understand Each Type of Research Question

Type	Purpose	Typical wording	Example
Descriptive	Describes *what exists* or *what is known*	“What are…”	“What biodegradable materials are used for food packaging?”
Comparative	Compares two or more entities	“How does X compare to Y…”	“How do biobased materials compare to conventional plastics in barrier performance?”

Type	Purpose	Typical wording	Example
Evaluative	Judges performance, suitability, or effectiveness	"To what extent is…" or "How effective is…"	"How suitable are biobased materials as sustainable packaging under EU PPWR criteria?"
Defining	Clarifies concepts or categories	"What defines…"	"What constitutes a biodegradable packaging material under EU law?"
Explaining	Identifies causes or mechanisms	"Why does…"	"Why do biobased films show higher oxygen permeability than PET?"
Designing	Proposes or creates new solutions	"How can we design…"	"How can we design biobased multilayer films for better moisture resistance?"

🧱 Step 3: Analyze Your Project Components

Component	Nature	Question Type
Building a FAIR dataset	**Descriptive** (collecting, mapping)	
Comparing materials vs. conventional plastics	**Comparative**	
Assessing suitability for sustainable food packaging	**Evaluative**	
Structuring metadata and open data publication	**Design-oriented** (a *technical design* element, but not design research)	

Step 4: Final Classification

Your **primary research question** is a mix of **descriptive**, **comparative**, and **evaluative**, but the **main focus** is **comparative with evaluative goals**.

In short:

> **Type:** *Comparative–Evaluative* (supported by descriptive data collection)

Example Framing for Your Thesis

Main Research Question (Comparative–Evaluative):

> "How do biobased and biodegradable packaging materials compare to conventional plastics in terms of physical, mechanical, and barrier properties, and how suitable are they for preserving food quality in line with EU sustainability goals?"

Supporting Descriptive Question:

> "What data on material properties and food preservation outcomes are available for biobased and biodegradable food packaging materials in the literature?"

🧱 Why This Matters

This classification helps you:

- Define your **methodology** (systematic review + data synthesis, not experimental testing).
- Justify your **analytical approach** (comparative tables, benchmarking, visualization).
- Clarify your **contribution** (an evaluative dataset for future design and policy use).

Summary:

Aspect	Your Thesis
Main question type	Comparative–Evaluative
Supporting type	Descriptive
Not included	Defining, explaining, designing (as research paradigms)
Method used	Systematic literature review + data curation + comparative analysis